

Final Report

MS&E 125

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EXECUTIVE SUMMARY

Gentrification has become one of the most important urban development challenges facing metropolitan regions throughout the United States. While neighborhood investment can improve economic conditions and public infrastructure, it often coincides with rising costs of living and the displacement of long-term residents. A goal of this project was to determine whether neighborhood- or “tract-level” data could be used to identify areas at elevated risk of future gentrification within the Bay Area. Using census tract data from San Francisco, Oakland, and San Jose, the study integrated a wide range of predictors, including demographics, household income, business density, and college education. Three different machine learning techniques were employed to compare model performances: Logistic Regression, Random Forest, and XGBoost. The models produced validated AUC scores of 0.587, 0.597, and 0.634, respectively. The averaged validated AUC of the 3 was 0.604. There were several important limitations to note, such as the small training set, data distribution shift with COVID, and ACS rolling average overlap. Each of these limitations can be addressed in one way or another and presents an excellent opportunity for future research. The biggest implication of this model was that it could be used to provide an early warning system to help cities proactively target these communities and allocate the necessary resources to effect change. A secondary consequence was the subsequent questions the model ultimately posed, considering how such information could be used and by whom, along with weighing the ethicality of gentrification.

INTRODUCTION

Gentrification is commonly defined as a process in which higher-income households, investors, and/or developers move into historically lower-income neighborhoods, leading to increases in housing costs, property values, and neighborhood desirability. The social consequences of gentrification have become increasingly visible in major urban regions such as the Bay Area. Rising rents and property values have contributed to displacement pressures, particularly among lower-income households and historically marginalized communities. As a result, policymakers increasingly seek tools that can identify neighborhoods experiencing early warning signs of gentrification. Therefore, the central research question guiding this project was:

Can neighborhood-level data from 2011–2024, including permit activity, business density, and demographic composition, predict which Bay Area census tracts are at the highest risk of gentrification in the near future, and which features are most responsible for those predictions?

The project outlined two objectives:

1. Develop a predictive model capable of identifying neighborhoods at elevated risk of gentrification.
2. Explain the factors driving risk within individual census tracts.

Building a comprehensive dataset proved to be a trickier task than first envisioned. The goal was to capture demographic, housing, labor market, and development characteristics that might serve as indicators of neighborhood change. There were 4 primary data sources used for the scope of the project, each outlining the counties of San Francisco, Oakland (Alameda County), and San Jose County:

1. American Community Survey (ACS)
2. Longitudinal Employer-Household Dynamics (LODES)
3. Building Permit Records
4. Zillow Home Value Index

The ACS provided a broad set of socioeconomic variables like household income, rent levels, home values, and educational attainment. Since we used a train-validation-test split, the ACS data consisted of rolling 5-year averages for 2015, 2020, and 2024. LODES supplied information regarding employment composition and job types associated with residents and workers in each tract. Building Permit Records had varying websites for each county. Once acquired, both residential and commercial new construction activity was pulled. The thought was that this new development would precede increases in housing prices and demographic turnover. Permit data was preferred over our previous amenity density metrics from Yelp precisely because permit activity is a leading indicator of capital investment. The Zillow Home Value Index contributed information regarding housing appreciation trends based on month-to-month figures.

FEATURE ENGINEERING/MODELING

A substantial portion of the project involved transforming raw data into meaningful predictive variables. All datasets were linked at the census tract level, creating a unified analytical dataset. This process allowed information from demographic, economic, and development sources to be analyzed simultaneously. Both absolute change and percentage change were calculated between successive

observation periods. The train data was pulled from 2011-2015 data. The validation data was pulled from 2016-2020 data. The test data was pulled from 2020-2024 data. According to the gentrification label set, all tracts that were below-median income and in the top quartile of growth in at least 3 out of 4 areas—household income, home value, rent, and college educational attainment—were selected. In regard to the midterm feedback, we wanted to ensure that the gentrification labeling occurred on relative scales, using the local city median income, and all features were calculated using within-period percentile rankings. In the finalized feature selection, a few key decisions were made:

- Zillow variables were ultimately excluded from the final feature set.
- City indicator variables ("city dummies") were added.
- Change metrics associated with variables used directly in the gentrification label were removed to prevent circular prediction.

Next, to reduce overfitting and eliminate redundant predictors, LASSO feature selection was employed. This method retained only the variables that contributed meaningful predictive information while shrinking less useful coefficients to zero. The project then compared three machine learning approaches:

1. Logistic Regression served as a baseline statistical model. Because of its interpretability and widespread use in classification tasks, it provided a useful benchmark for evaluating more complex methods.
2. Random Forest uses an ensemble of decision trees to capture nonlinear relationships and interactions among predictors. This approach often performs well when relationships are complex and difficult to model with linear techniques.
3. XGBoost is a gradient boosting algorithm that iteratively improves prediction performance by correcting previous errors. The method is widely recognized for achieving strong results on structured tabular datasets.

Among the three models evaluated, XGBoost produced the highest validation performance while also resulting in the smallest cross-validated standard deviation. The final model chosen was an ensemble of the three. We chose to report the risk scores determined by our ensemble model to minimize the individual biases of each of the three different models. Probabilities generated by Logistic Regression, Random Forest, and XGBoost were averaged to produce a final risk score ranging from 0 to 1. Refer to figures 1 and 2 for the output results map. The rationale behind this approach is that different models capture different aspects of the data. Combining them can create more robust predictions than relying on a single algorithm.

RESULTS

Model performance was evaluated using Area Under the Receiver Operating Characteristic Curve (AUC), a common metric for binary classification.

Model	Val AUC	CV AUC	CV STD
Logistic Regression	0.587	0.693	0.176 (+ or -)
Random Forest	0.597	0.582	0.261 (+ or -)
XGBoost	0.634	0.793	0.159 (+ or -)
3-model ensemble	0.604	-	-

XGBoost achieved the highest validation AUC of 0.634 and the strongest cross-validation performance, making it the preferred model. Although these performance levels are moderate rather than exceptionally high, they suggest that neighborhood-level indicators contain meaningful predictive information regarding future gentrification risk. Using SHAP (Shapley Additive Explanations), the variables contributing most strongly to model predictions, outside of the gentrification qualification labels, were:

1. Change in the African American population (decrease))
2. Overall population size (increase)
3. Share of jobs in the food industry (increase)
4. Share of housing built before 1940 (increase)
5. Share of residents aged 25–34 (increase)

These variables align with patterns frequently discussed in urban economics literature. Younger adult populations, changing demographic composition, older housing stock, and shifts in local economic activity often characterize neighborhoods undergoing redevelopment and socioeconomic transition.

The midterm feedback noted that for a city planner audience, interpretability often matters more than raw AUC, which often leads to logistic regression being the most relevant model. However, for a city planner, they care about the most tangible features, and giving them the SHAP values extends this interpretability to the XGBoost model, telling the city planner which features they should target their efforts toward.

Figure 1. Overview of output results, highlighting the regions at the highest risk of future gentrification.

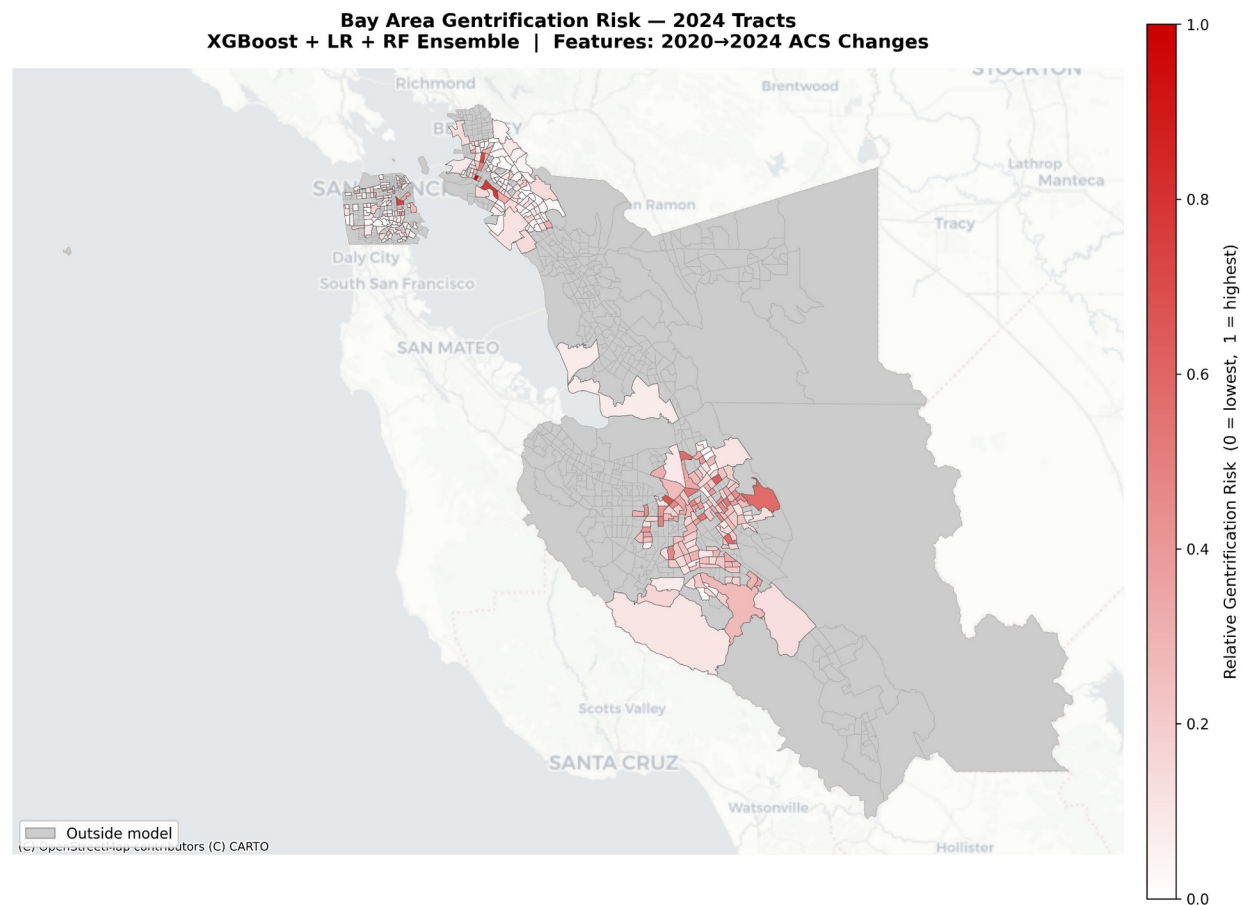
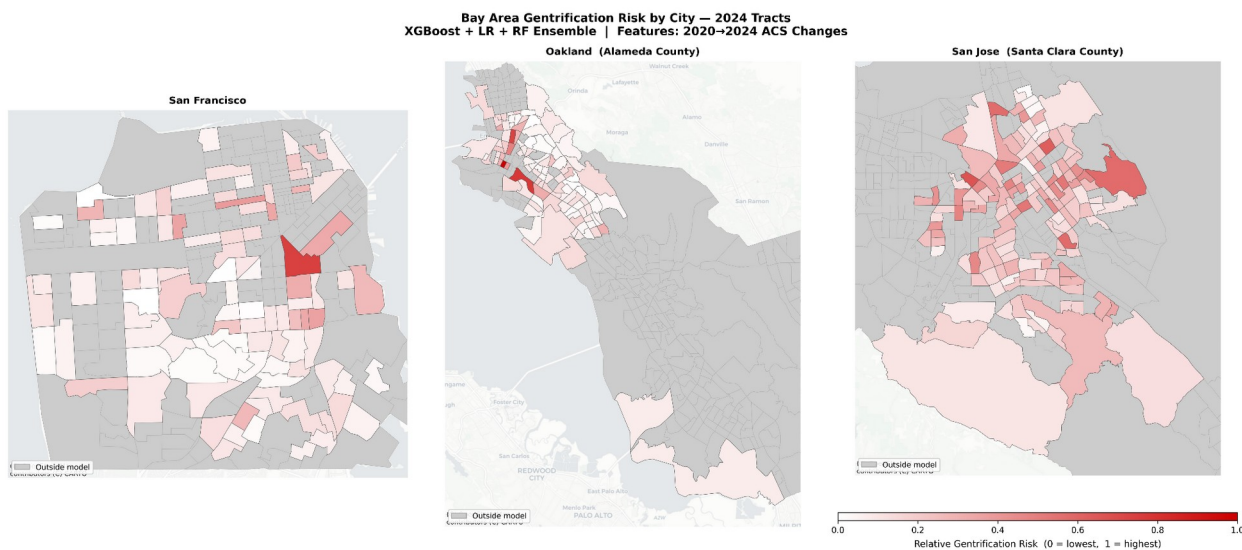


Figure 2. Individual county map outputs identifying future gentrification risks.



LIMITATIONS

It is also important to note the limitations of the model and what was done to either overcome the limitations or what would need to be addressed in future studies.

1. **Limited Positive Cases:** Gentrification events are relatively rare compared to non-gentrifying neighborhoods. To address this imbalance, the study used SMOTE (Synthetic Minority Oversampling Technique) to increase the representation of positive cases during training. While SMOTE improves predictive performance, it can also increase model variance and instability. Another solution would have been to expand the geographic range of the study, but that prompted problems in terms of locating permit information and building density.
2. **Market Regime Shifts:** The training data primarily reflected pre-COVID conditions, whereas testing covered post-COVID periods. Because housing markets and migration patterns changed significantly during and after the pandemic, model performance may be affected by these shifting dynamics.
3. **ACS Rolling Average Overlap:** The ACS uses five-year rolling averages, creating overlap between validation and testing periods since the testing data was on the most recent data released: 2024. The argument was that this does not create a large data leakage because the individual overlapping years are being hidden with the larger averages of their separate 5-year time spans. Even so, the shared years reduce complete temporal independence between datasets.
4. **Spatial lag variables (PySAL):** The midterm feedback flagged that gentrification doesn't happen in a vacuum, and nearby tracts are also affected when wealth is brought into an area. In future research, it would be important to implement PySAL spatial lag features to model that geographic dependency more so than just comparing tracts within their city, but specifying one tract's effect on tracts nearby.

POLICY IMPLICATIONS/ETHICAL CONSIDERATIONS

One of the project's most important contributions is its potential policy application and the ethical considerations it brings up. The model functions as an early warning system, allowing governments and nonprofit organizations to identify neighborhoods experiencing early signs of gentrification. Such information could support interventions including:

- Affordable housing preservation
- Tenant protection programs
- Community land trusts
- Anti-displacement initiatives

However, there is also an ethical concern to be acknowledged. Predictive information can be used in different ways depending on the user accessing it. While policymakers may use risk scores to protect vulnerable communities, investors and developers could use the same information to identify profitable opportunities. This creates the possibility of a feedback loop in which predictions themselves accelerate the very process being predicted. The midterm feedback proposed building a Directed Acyclic Graph to clarify whether amenities are causes or downstream effects of gentrification, which relates to this idea of feedback loops and must be understood before recommending specific policy adjustments.

Neighborhoods flagged as "high risk" may attract speculative investment, potentially contributing to increased displacement pressures. This brought up more questions, such as:

1. Should information, such as a map identifying future gentrification risk of tracts, be made available to the public?
2. If gentrification is bad, or at least has the potential to be, then what is the effective way to introduce resources and encourage upward mobility in communities without creating these negative externalities?

These are the questions that should be asked in order to prompt conversations of change.

Lastly, there were thoughts on what future research could entail or how this research would be expanded. There were two main ideas around expansion: increasing sample size and choosing a different geographic region. Sample size was a notable limitation in the study that would have increased the robustness. In addition, there was an awareness that the Bay Area is a higher cost of living area compared to most of the United States. It would be interesting to see this model replicated in other cities that are more prone to gentrification, such as Chicago or Atlanta.

CONCLUSION

This project demonstrates that neighborhood-level demographic, economic, employment, and development indicators can provide meaningful insight into future gentrification risk across Bay Area census tracts. By integrating ACS, LODES, permit, and housing market data, a predictive framework was constructed that was capable of identifying neighborhoods likely to undergo substantial socioeconomic transformation. Among the evaluated models, XGBoost delivered the strongest predictive performance, while SHAP analysis provided transparency into the factors driving risk predictions. Although limitations remained—including small sample sizes, shifting market conditions, and the challenges of defining gentrification—the project offered a promising approach for proactive urban planning.

Most importantly, the model illustrates how predictive analytics can be used not merely to forecast neighborhood change but to inform policy decisions aimed at reducing displacement and preserving community stability. At the same time, the project highlights the ethical responsibility required when deploying predictive tools, since the same information that helps policymakers protect residents could also be used by market participants in ways that reinforce gentrification pressures.